

JUNXIAN WU

Forward-Deployed AI Engineer | Computer Vision & Multimodal Systems

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Target roles: Forward-Deployed AI Engineer | Applied AI Engineer | Computer Vision ML Engineer | Multimodal AI Engineer | ML Systems Engineer

PROFILE

Singapore-based AI engineer with 4+ years building deployable computer-vision and multimodal systems for camera and image workflows. Strongest at the intersection of model development, edge deployment, C++/Android runtime integration, computational photography, data/evaluation loops, and customer-facing productization.

Best fit for Forward-Deployed AI / Applied AI roles where model quality is only one part of the problem: the system also needs discovery, technical scoping, deployment constraints, visual-quality debugging, acceptance gates, and maintainable handoff.

CORE STRENGTHS

Forward-deployed AI: technical discovery, solution shaping, prototype-to-deployment delivery, product/customer feedback translation, issue triage, acceptance criteria, and reusable handoff patterns

Computer vision: segmentation, image matting, monocular depth, object detection, optical-flow tracking, camera AI, computational photography, and visual quality debugging

Model deployment: PyTorch-to-TFLite/ONNX export, QAT/PTQ-style workflows, INT8 validation, TensorRT/ONNXRuntime, C++/Android runtime integration, and latency/accuracy trade-offs

Data and evaluation: hard-case taxonomy, dataset gates, model comparison reports, golden cases, visual montages, SOPs, and model replacement checks

Multimodal systems: VLM/LLM workflows, structured outputs, tool calling, multi-stage orchestration, provider comparison, evaluation hooks, and reproducible traces

PROFESSIONAL EXPERIENCE

Black Sesame Technologies (Singapore) Pte Ltd | Singapore

Senior AI Engineer | Jan 2025 - Present

- Owned forward-deployed camera and image AI productization: translated visual-quality feedback into reproducible issue sets, failure taxonomies, data/model iteration plans, acceptance gates, and roadmap input.
- Organized AI composition delivery as a system rather than a rule list: detector exploration, visual debugging tools, product/algorithm review cadence, real-device validation, issue triage, and handoff documentation.
- Drove mask/matting quality work across hard-case data, hair and edge behavior, foreground objects, middle-ground selection, model replacement evaluation, and data-driven training SOPs.
- Designed VLM-based image-composition workflows with structured outputs, multi-stage orchestration, provider comparison, retry/fallback logic, review rubrics, and reproducible debug traces.
- Integrated deep models with GPU rendering and service pipelines using OpenGL/GLSL, TensorRT, Docker, FastAPI/Flask, ONNXRuntime, and Python/C++ tooling.

AI Engineer | Aug 2022 - Jan 2025

- Built and maintained a reusable vision model training-to-deployment foundation for segmentation, matting, depth, and camera AI tasks, connecting data processing, training, evaluation, quantization, export, deployment validation, and debugging.
- Built C++/Android simulation and deployment workflows for camera AI modules, covering CMake/native builds, Android NDK integration, adb-based device workflows, runtime libraries, and production reference documentation.
- Worked on depth-aware computational photography, combining monocular depth, portrait segmentation/matting, CPU/GPU rendering, subject protection, highlight/spot behavior, edge transition control, and device-aware tuning.
- Led native real-time composition work combining TFLite object detection, optical-flow tracking, camera-pose guidance, and state-machine decision logic for live camera interaction.
- Developed data and evaluation infrastructure for production readiness, including hard-case data requirements, device dump extraction, image-format conversion, batch simulation, visual comparison reports, and reproducible review packages.
- Built model conversion and deployment utilities across PyTorch, TensorFlow/Keras bridge code, TFLite, ONNX, INT8/FP16-style deployment paths, Dockerized inference, and REST-based model serving.

National University of Singapore, Safety and Resilience Research Unit | Singapore

Research Intern, Part-time | Dec 2021 - Jul 2022

- Developed computer-vision methods for construction-site hazard and clutter classification, including dataset preparation, labeling, model experimentation, and spatial risk analysis.
- Co-authored a peer-reviewed Automation in Construction paper on supervised and self-supervised feature representation learning for cluttered construction housekeeping image classification.

Shandong University | China

Research Assistant, Part-time | Sep 2019 - Jun 2021

- Applied computer vision and deep learning to cross-disciplinary psychology research, building datasets and analyzing stereotypical motions of children with autism spectrum disorder to support early-diagnosis studies.

TECHNICAL LEADERSHIP LINES

- Vision model platform and deployment foundation: dataset gates, training configs, loss/model variants, quantization/export paths, validation checks, and downstream runtime constraints.
- On-device vision systems: C++/Android integration, TFLite runtime, detection/tracking, state machines, runtime fallbacks, device-side debugging, and cross-platform builds.
- Computational photography and rendering: depth, segmentation, matting, alpha compositing, AI bokeh, highlight/spot behavior, OpenGL/GLSL rendering, and TensorRT/ONNXRuntime services.
- Data, evaluation, and handoff: hard-case mining, model comparison, visual montages, golden cases, SOPs, release checks, and production handoff documentation.
- Agentic multimodal workflows: explicit workflow state, structured outputs, provider comparison, evaluation harnesses, cost/quality trade-offs, and reproducible run artifacts.

EDUCATION

National University of Singapore: Master of Technology in Intelligent Systems, GPA 4.33/5.0, Jul 2021 - Jun 2022

Shandong University: Bachelor of Software Engineering, GPA 86.86/100, Sep 2017 - Jun 2021; Outstanding Graduate

Oxford Prospects Programme, Regent's Park College: Overseas Experience Scholarship, Aug 2020

SELECTED KEYWORDS

Forward-Deployed AI, Applied AI, Computer Vision, Multimodal AI, Vision-Language Models, Production ML, Model Deployment, Camera AI, TFLite, ONNX, TensorRT, ONNXRuntime, PyTorch, Python, C++, OpenCV, OpenGL, GLSL, Docker, FastAPI, Android NDK, Segmentation, Image Matting, Monocular Depth, Object Detection, Optical Flow, Computational Photography, Structured Outputs, Evaluation.